

Ebuka Philip Oguchi Ph.D.

CONTACT INFORMATION

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ACADEMIC APPOINTMENT

Assistant Professor (Tenure Track)
Department of Computer and Information Sciences, Spelman College
January 2026 – Present

RESEARCH INTERESTS

Cybersecurity and trust establishment for emerging wireless systems; message integrity and authentication; physical-layer security; location and RF fingerprinting; autonomous and vehicular networks; agricultural IoT; molecular and nano-scale communication security; applied machine learning for adversarial environments.

EDUCATION

- **University of Nebraska–Lincoln**
Ph.D. in Computer Science and Engineering, 2025
Advisor: Prof. Nirnimesh Ghose
Dissertation: *Authentication and Message Integrity Verification for Emerging Wireless Networks*
- **Changchun University of Science and Technology, China**
M.S. in Computer Applied Technology, 2020
Advisor: Prof. Fang Ming
Thesis: *Smoke Recognition Using ResNet and GoogleNet*
- **Nnamdi Azikiwe University, Awka, Nigeria**
B.Eng. in Electronic and Computer Engineering, 2016
Project: *Biometric Course Register Using Fingerprint Authentication*

PUBLICATIONS

Journal Articles (Under Review)

1. Oguchi, E.; Ghose, N.; Vuran, M.C. *Soil-Assisted Trust Establishment for Underground Internet-of-Things*. IEEE Transactions on Wireless Communications.
2. Oguchi, E.; Anderson, M.; Duong, T.T.; Wisniewska, A.; Ghose, N. *Systematization of Knowledge for Security in Molecular and Nano-Communications*. IEEE Transactions on Molecular, Biological, and Multi-Scale Communications.

Peer-Reviewed Conference Papers

1. Oguchi, E.; Ghose, N.; Vuran, M.C. *STUN: Secret-Free Trust Establishment for Underground Wireless Networks*. IEEE INFOCOM Workshops, 2022.
2. Oguchi, E.; Ghose, N.; Vuran, M.C. *VET: Autonomous Vehicular Credential Verification Using Trajectory and Motion Vectors*. EAI SecureComm, 2023.

RESEARCH EXPERIENCE	Research Assistant, University of Nebraska–Lincoln 2021–2025 • Agricultural IoT Security: Secure bootstrapping and authentication for underground sensor networks using physical-layer features. • Autonomous Vehicle Security: Message integrity verification using trajectory, motion vectors, and Doppler effects. • RF Fingerprinting: Machine-learning-based authentication for underground and OTA environments. • Molecular Communication Security: Lightweight security protocols for nano-scale biological communication.
RESEARCH FUNDING (AS PART OF FUNDED PROJECTS)	• Research conducted as part of NSF-funded projects (CNS-2331191, CNS-2225161, CNS-1619285, ECCS-2030272). • Research conducted as part of projects supported by the Nebraska Center for Energy Sciences Research (NCESR).
TEACHING EXPERIENCE	• Assistant Professor, Spelman College Spring 2026 (Courses TBA) Department of Computer and Information Sciences Expected to teach undergraduate computing courses beginning Spring 2026. Course assignments will be finalized by the department. • Teaching Assistant, Cryptography & Security (CSCE 477/877) Fall 2022 University of Nebraska–Lincoln Supported labs and assignments on block ciphers, RSA, hash functions, and network security protocols. Helped students implement cryptographic primitives and analyze protocol correctness. • Teaching Assistant, Introduction to Computer Architecture Jan 2018 – May 2018 D.S. Adegbenro ICT Polytechnic Delivered lectures, supervised labs, graded homework/exams, and supported foundational computer systems instruction. • Teaching Assistant, Introduction to Computer Software Jan 2018 – May 2018 D.S. Adegbenro ICT Polytechnic Taught introductory programming concepts, organized practical sessions, and supervised assessments.
PROFESSIONAL SERVICE	• Reviewer: IEEE Transactions on Industrial Informatics; ACM Transactions on Cyber-Physical Systems. • Reviewer: EAI SecureComm; ACM WiSec. • Judge: UNL College of Engineering Poster Fair.
MENTORSHIP	• Google CS Research Mentorship Program (CSRMP). • Institute for African-American Mentoring in Computing Sciences (iAAMCS). • Mentored Ph.D. and undergraduate researchers at UNL.

AWARDS

- Ph.D. Dissertation Award, University of Nebraska–Lincoln, 2025.
- Mary E. and Elmer H. Dohrmann Fellowship, 2023.
- Chinese Government Scholarship, 2018.

SKILLS

Machine learning (TensorFlow); Python, MATLAB, C; cryptographic protocol analysis; SDRs (USRP, GNU Radio); network security; incident response; Wireshark; adversarial ML; hardware and wireless experimentation.